Original article

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Association between plasma interleukin-17 levels and risk of psoriasis: a meta-analysis

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Summary

Background. Emerging evidence indicates that plasma interleukin (IL)-17 levels may be associated with increased risk of psoriasis, but the individual published results are inconclusive.

Aim. To evaluate IL-17 levels in patients with psoriasis using a meta-analysis of studies comparing IL-17 levels in controls and in patients with psoriasis.

Methods. All relevant studies were identified by searching PubMed, Web of Science and MEDLINE databases before 1 November 2015. Pooled risk estimates were calculated by random-effects models. Crude OR and standardized mean difference (SMD) with corresponding 95% CI were also calculated.

Results. In total, eight cross-sectional study studies were included in the final analysis. The mean plasma levels of IL-17 were higher in patients with psoriasis than in healthy controls (SMD = 0.47, 95% CI = 0.07–0.86, $P < 0.001$).

Conclusions. Based on this literature review, there appears to be a positive association between IL-17 plasma levels and psoriasis.

Introduction

Psoriasis is a cutaneous inflammatory disease that affects 2–5% of the population worldwide.[1] The typical presentation of psoriasis includes erythematous and silvery-white scaly skin plaques, found mainly on extensor areas, and there can also be dystrophic changes in the nails. Based on the the latest literature, psoriasis is now considered a chronic, systemic inflammatory disorder. It also appears to be a variably dominant, genetically transmitted disease of multifactorial aetiology.[2]

Interleukin (IL)-17 is a cytokine produced by mast cells, innate lymphoid cells (ILCs), unconventional T cells and typical innate immune cells, such as neutrophils and macrophages in the epithelial barrier. IL-17 can respond to infectious agents rapidly by recruiting neutrophils as the first defensive line and triggering the production of antimicrobial peptides.[3] T helper (Th)17 cells differentiate from CD4+ T cells in the presence of transforming growth factor (TGF)-β and IL-6, and are the major source of IL-17. Recently, there has been a growing understanding of the significance of the Th17 subset of T cells and their relevant cytokines, including IL-17, both in the biological mechanism of innate host defence and in the pathogenesis of autoimmune or inflammatory diseases, such as psoriasis and psoriatic arthritis.[4] In addition, recent evidence has indicated that there may be an association between IL-17 plasma levels and increased psoriasis risk, but the individual published results are inconclusive.

In the current study, we performed a meta-analysis with all eligible case–control studies to obtain insights into these associations. This meta-analysis may promote understanding of the precise role of IL-17 in the progression of psoriasis.

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Conflict of interest: the authors declare that they have no conflicts of interest.

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Methods

Search strategy

Three investigators (ZX, BJ, HB) independently searched published studies indexed in PubMed, Web of

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Science and MEDLINE from their inception to 1 November 2015, using the search terms 'psoriasis', 'IL-17' and 'interleukin-17'. Details of our search strategy are described in Supplementary data online. A manual search of the references of the retrieved articles meeting the inclusion criteria was also performed.

Inclusion criteria

All of the included studies had to meet all of the following four criteria: (i) study design: clinical cohort, case–control study or cross-sectional study; (ii) study subject: relationship between IL-17 plasma levels and psoriasis; (iii) patients: had to meet the diagnostic criteria for psoriasis; and (iv) inclusion of IL-17 plasma levels.

The following data were extracted: first author's name, year of publication, total number of cases and controls, age and ethnicity of participants, IL-17 plasma levels of participants, psoriasis type, measurement method and blood sample type. The three investigators extracted all data independently, complied with the selection criteria, and reached a consensus on all items.

Statistical analysis

Crude OR or standardized mean difference (SMD) with 95% CI were calculated. Statistical significance of the pooled estimate was examined using the Z test. Power calculations were performed using PS (Power and Sample Size Calculations) software.[1] Between-study variations and heterogeneities were estimated using Cochran Q statistic, with $P < 0.05$ was set as the cut-off to identify heterogeneity in the associations.[1] We also quantified the effects of heterogeneity by using the I^2 test (ranging from 0–100%), which represents the proportion of interstudy variability that can be contributed to heterogeneity rather than to chance. When a significant Q test with $P < 0.05$ or $I^2 > 50\%$ indicated the existence of heterogeneity between studies, a random-effects model was conducted for the meta-analysis; otherwise, a fixed-effects model was used. To explore potential sources of heterogeneity, subgroup analyses were performed based on ethnicity and disease type. Sensitivity analysis was performed by omitting each study in turn to assess the quality and consistency of the results. Begger funnel plots and Egger linear regression test were used to evaluate publication bias.[5] Two-sided $P < 0.05$ was considered statistically significant. All calculations were performed using STATA software (v12.0; STATA Corp., College Station, TX, USA).

Results

In total, 164 studies matching the search terms were identified. Based on the inclusion criteria, 9 case–

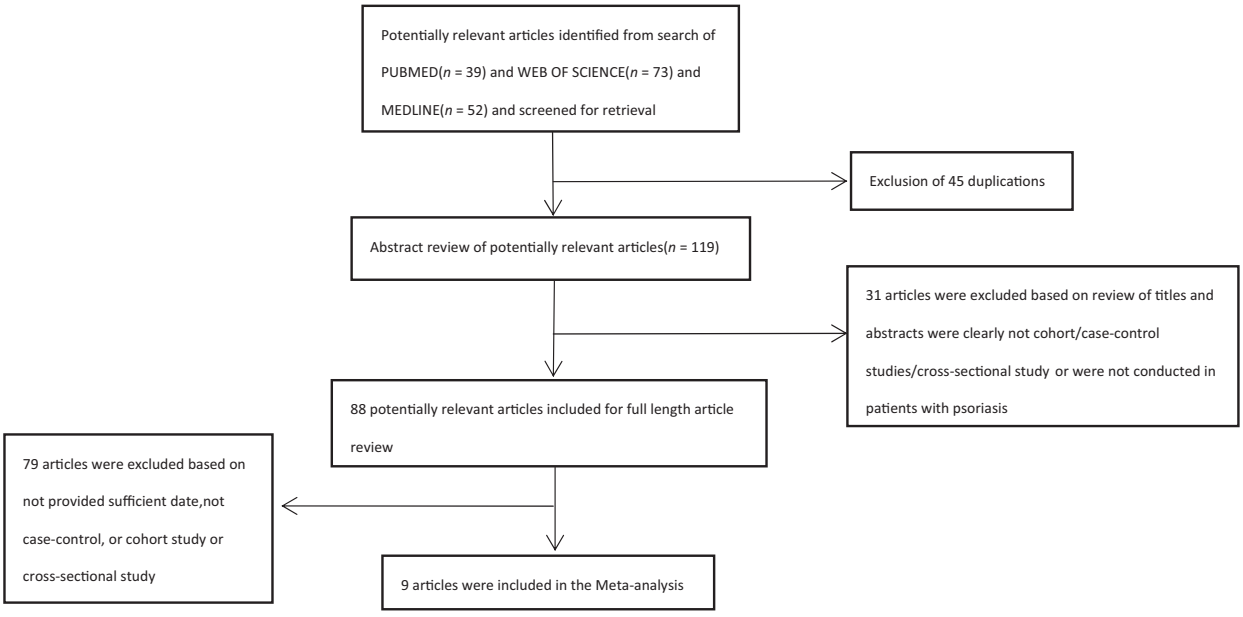

Figure 1 Flow chart showing study selection procedure. Nine cross-sectional study were included in this meta-analysis and 155 studies were excluded.

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control studies[1,6–13] were initially included in this meta-analysis, and the remaining 155 were excluded. The flowchart of study selection is shown in Fig. 1, and the characteristics of the included studies are shown in Table 1.

The 9 studies comprised 681 participants [345 patients with psoriasis, 234 healthy controls (HCs)] who met the inclusion criteria. The publication year of the included studies ranged from 2005 to 2015. Three studies were conducted in European populations and six studies in Asian populations.

Association between IL-17 plasma levels and psoriasis risk

All nine studies examined the association between the mean levels of IL-17 plasma and psoriasis. Because heterogeneity existed ($I^2 = 84.4$), a random-effects model was used. The meta-analysis results revealed that the study from Xuan *et al.* on the mean levels of IL-17 plasma in patients with psoriasis and HCs was an outlier (Fig. S1). Upon further analysis, we found that the measurement method used in that study was flow cytometry while the other studies used ELISA, and this difference in methods might explain the difference in results. We therefore excluded this article, leaving us with eight studies on the association between the mean levels of IL-17 plasma and psoriasis for the final analysis.

Because there was obvious heterogeneity between the studies ($I^2 = 78.7$, $P < 0.001$), a random-effects model was used. The meta-analysis results revealed that the mean plasma levels of IL-17 in patients with psoriasis were higher than those in HCs (SMD = 0.47, 95% CI 0.07–0.86, $P < 0.001$) (Fig. 2). The results of subgroup analysis by ethnicity revealed that there was no significant association with IL-17 plasma levels in either the European (SMD = 0.52, 95% CI −0.08 to 1.11, $P = 0.03$) or the Asian (SMD = 0.44, 95% CI −0.14 to 1.01, $P < 0.001$) populations studied.

Sensitivity analysis and publication bias

Sensitivity analysis was performed to assess the influence of each individual study on the pooled estimates by omitting individual studies in turn. The results suggested that no individual study significantly affected the pooled estimates for the associations of plasma IL-17 levels with psoriasis risk, and that the results were stable and reliable (Fig. 3).

An Egger linear regression test was performed to assess the publication bias of included studies. The shapes of the funnel plots revealed that the data from

Table 1 Characteristics of studies included in this meta-analysis.

Reference	Ethnicity	Number of patients		Age, years		Plasma level of IL-17		Psoriasis type	Sample type	Measurement method	PASI
		Cases	Controls	Cases	Controls	Cases	Controls				
Caproni *et al.*, 2008[1]	European	30	10	–	–	52.86 ± 27.63	19.68 ± 9.78	PV	Blood	ELISA	–
Murakami *et al.*, 2011[6]	Asian	48	20	–	–	613.80 ± 603.20	5.60 ± 15.51	PP	Blood	ELISA	–
Zhang *et al.*, 2009[7]	Asian	54	18	–	–	62.79 ± 13.63	50.66 ± 12.86	–	Blood	ELISA	–
Arican *et al.*, 2005[8]	Asian	30	23	35.00 ± 15.50	–	8.30 ± 3.80	7.40 ± 3.76	PV	Blood	ELISA	9.3 ± 8.15
Kanda *et al.*, 2010[9]	Asian	61	31	51.90 ± 20.90	46.4 ± 15.00	30.00 ± 8.50	17.00 ± 50.60	PV	Blood	ELISA	8.09 ± 6.83
Yilmaz SB *et al.*, 2012[10]	Asian	30	50	39.90 ± 14.90	39.65 ± 13.40	6.20 ± 5.60	4.40 ± 4.10	PV	Blood	ELISA	6.60 ± 5.40
Michalak-Stoma A *et al.*, 2013[11]	European	60	30	45.60 ± 13.20	–	3.16 ± 1.83	2.86 ± 1.49	–	Blood	ELISA	15.7 ± 9.70
Kyriakou A *et al.*, 2014[12]	European	32	32	44.53 ± 15.60	–	2.67 ± 3.46	1.80 ± 2.81	PV	Blood	ELISA	–
Xuan ML *et al.*, 2015[13]	Asian	62	20	–	37.25 ± 12.78	7.44 ± 7.58	37.04 ± 71.24	PV	Blood	FC	12.84 ± 7.60

ELISA, enzyme-linked immunosorbent assay; FC, flow cytometry; IL, interleukin; PASI, Psoriasis Area and Severity Index; PP, pustular psoriasis; PV, psoriasis vulgaris.

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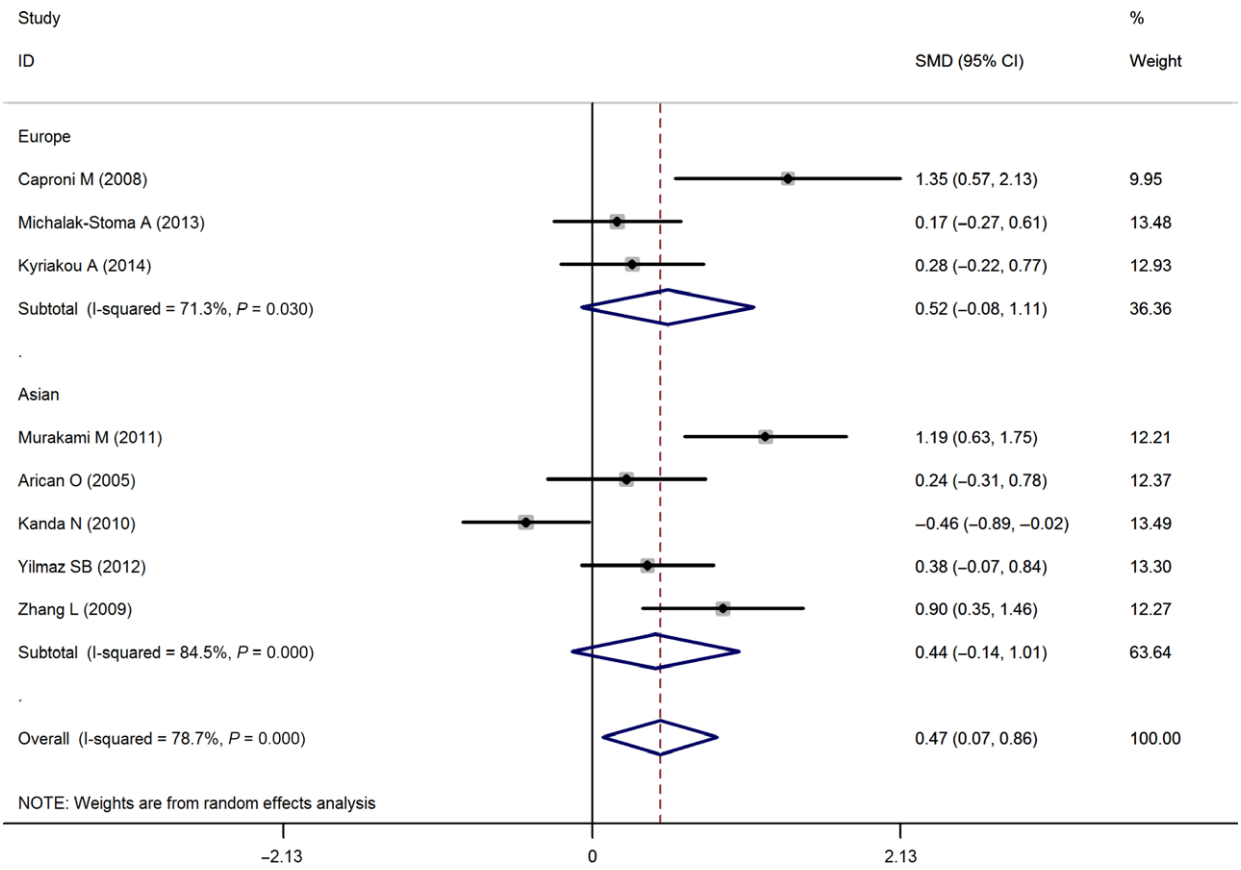

Figure 2 Standard mean difference and pooled estimates for interleukin (IL)-17 levels in patients with psoriasis and healthy controls.

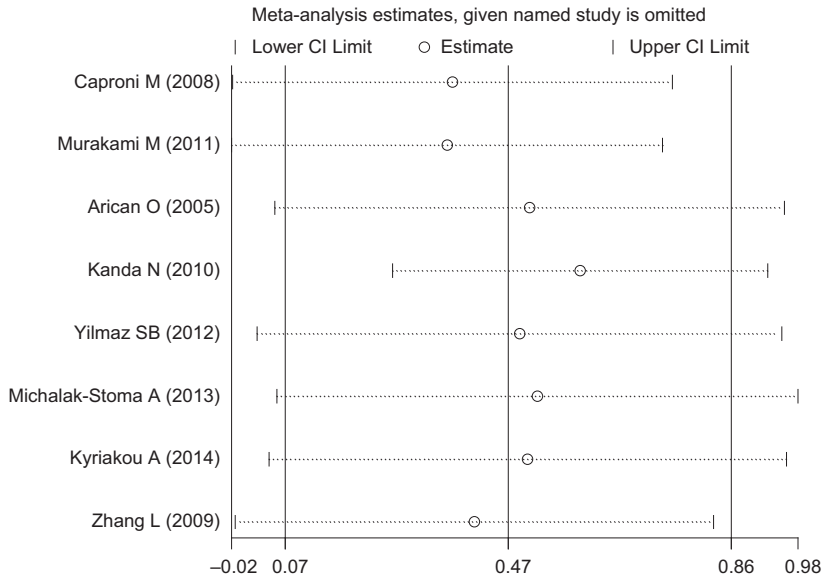

Figure 3 Influence on adjusted effect estimates of removing individual studies. Circles are effect estimates, horizontal dotted lines are 95% CIs for meta-analysis of the remaining studies, and the vertical line in the centre is the pooled effect estimate for all studies.

Kanda *et al.* showed some evidence of asymmetry for the associations of plasma IL-17 levels with psoriasis risk (Fig. S3). This may have been due to their small sample size. The Egger test displayed strong statistical evidence ($t = 2.94$, $P = 0.0$) for publication bias in all of the studies.

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Discussion

Psoriasis is a common cutaneous disease whose main characteristic is the excessive turnover of skin cells, while activated Th cells are considered to be among its major pathogenic modulators.[14] Most of the evidence for IL-17 produced by innate immune cells in the perpetuation of autoimmune and chronic inflammatory disorders originates from rheumatoid arthritis[15] and psoriatic disease (which includes both psoriatic arthritis and skin psoriasis[16,17]), two complex conditions in which IL-17 targeting is at an advanced stage of development. However, the mechanisms of how plasma levels of IL-17 raise the risk of psoriasis are not yet clear. Therefore, we conducted a meta-analysis of all suitable studies to obtain a more accurate estimate of these particular associations. The aim of the analysis was to study the correlation between plasma IL-17 levels and psoriasis.

This study is, to our knowledge, the first meta-analysis to centre upon the associations between plasma IL-17 levels and psoriasis. The meta-analysis included 8 qualifying studies, involving 345 patients with psoriasis and 234 HCs. We found a significant difference in mean plasma level of IL-17 between patients with psoriasis and HCs. Our conclusions are consistent with some previous studies, indicating that plasma levels of IL-17 may be associated with psoriasis.

This meta-analysis has several limitations. First, only eight articles fulfilled the criteria to be included in this meta-analysis. Second, we found during analysis that many of the articles used line charts and scatter plots to present the plasma levels of IL-17, without providing mean values and standard deviations, so that we could not extract more data from these articles. Therefore, the sample size was comparatively small and may not have had adequate statistical data to estimate the association between IL-17 and psoriasis risk. Thus, more studies with bigger sample sizes are needed for better statistical analysis. Third, because a meta-analysis is a cross-sectional study, it may include selection bias, which may have influenced the reliability of our findings.[18] Fourth, the statistical heterogeneity was high for the whole analysis. Of the nine articles we chose initially, we found that several did not classify by type or severity of psoriasis, which had a negative impact on our subgroup analysis to find the sources of the heterogeneity. Consequently, detailed studies are still needed to support our conclusions. However, based on the sensitivity analyses, our results were robust.

Conclusion

This study demonstrates that IL-17 level is upregulated in patients with psoriasis, which suggests that IL-17 may be involved in the development of psoriasis. Because of the aforementioned limitations to our study, further studies, both epidemiological and experimental, are warranted to clarify the association between IL-17 and the development of psoriasis, and to confirm our results.

What's already known about this topic?

- IL-17 may be involved in the aetiology of psoriasis.
- The association between IL-17 and psoriasis risk is still ambiguous.

What does this study add?

- Mean plasma levels of IL-17 were higher in patients with psoriasis than in HCs.
- Overall, increased risk of psoriasis appears to be associated with IL-17.

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